

Bijections and Permutations

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Subsets

Let A be a set with n elements.

We proved that A has 2^n subsets by deciding, for each element, whether to include it.

We now prove the same result in a different way: we represent every subset by a string of zeros and ones.

Bit strings

A **bit** is either 0 or 1.

A **bit string of length** n is an ordered sequence

$$b_1 b_2 \cdots b_n, \quad b_i \in \{0, 1\} \text{ for } 1 \leq i \leq n.$$

For example, 101 and 000 are bit strings of length 3.

Each of the n positions has two possible values, so the product rule gives

$$2^n$$

bit strings of length n .

Representing a subset

Fix the order a, b, c for the elements of

$$A = \{a, b, c\}.$$

For a subset $S \subseteq A$, write a 1 in the position of an element that belongs to S , and write a 0 otherwise.

Thus

$$\{a, c\} \longleftrightarrow 101, \quad \{b\} \longleftrightarrow 010, \quad \emptyset \longleftrightarrow 000.$$

The fixed order is essential: it tells us which element each position represents.

All subsets of $\{a, b, c\}$

The representation pairs every subset with one bit string:

$\emptyset$	$\longleftrightarrow$	000
$\{c\}$	$\longleftrightarrow$	001
$\{b\}$	$\longleftrightarrow$	010
$\{b, c\}$	$\longleftrightarrow$	011
$\{a\}$	$\longleftrightarrow$	100
$\{a, c\}$	$\longleftrightarrow$	101
$\{a, b\}$	$\longleftrightarrow$	110
$\{a, b, c\}$	$\longleftrightarrow$	111

One-to-one correspondence

A **one-to-one correspondence**, or **bijection**, between two sets pairs their elements so that

- ▶ every element of the first set is paired with exactly one element of the second set, and
- ▶ every element of the second set is paired with exactly one element of the first set.

Two finite sets connected by a bijection have the same cardinality.

A bijection therefore lets us count one collection by counting another.

Subsets and bit strings

Let

$$A = \{a_1, a_2, \dots, a_n\}.$$

Associate a subset $S \subseteq A$ with the bit string $b_1 b_2 \cdots b_n$, where

$$b_i = \begin{cases} 1, & a_i \in S, \\ 0, & a_i \notin S. \end{cases}$$

Every subset produces exactly one bit string, and every bit string determines exactly one subset. This is a bijection.

Hence

$$|\{S : S \subseteq A\}| = |\{\text{bit strings of length } n\}| = 2^n.$$

Two proofs of the same theorem

We now have two proofs that an n -element set has 2^n subsets.

The first proof constructs a subset through n include-or-exclude choices.

The second proof constructs a bijection between subsets and bit strings.

The second argument gives a reusable principle:

*To count a difficult collection, pair it bijectively
with a collection that is easier to count.*

Product rule for several choices

Suppose an object is constructed through k successive choices. If choice i has n_i possibilities for every result of the preceding choices, then the number of objects is

$$n_1 n_2 \cdots n_k.$$

This is the product rule applied repeatedly. The phrase “for every result of the preceding choices” matters: the number n_i must not depend on which earlier choices were made.

License numbers

A license number consists of two English letters followed by an integer from 1 to 9999. Repeated letters are allowed.

There are

- ▶ 26 choices for the first letter,
- ▶ 26 choices for the second letter, and
- ▶ 9999 choices for the integer.

By the product rule, the number of license numbers is

$$26^2 \cdot 9999.$$

Ice-cream flavors

Ten students stand in a line. Each student receives one of four flavors, and adjacent students must receive different flavors.

The first student has 4 choices.

Each later student has 3 choices because only the flavor given to the preceding student is excluded.

Therefore the number of assignments is

$$4 \cdot 3^9.$$

Permutations

A **permutation** of n distinct objects is an ordering of all n objects.
For example, the permutations of $\{a, b, c\}$ are

$abc, acb, bac, bca, cab, cba.$

Thus three distinct objects have six permutations.
The word “distinct” matters: each object can be identified separately.

Counting permutations

To construct a permutation of n distinct objects,

- ▶ choose one of n objects for the first position,
- ▶ choose one of the remaining $n - 1$ objects for the second position,
- ▶ continue until one object remains for the last position.

The product rule gives

$$n(n - 1)(n - 2) \cdots 2 \cdot 1.$$

Factorial

For an integer $n \geq 1$, define

$$n! = n(n-1)(n-2) \cdots 2 \cdot 1.$$

We also define

$$0! = 1.$$

Therefore the number of permutations of n distinct objects is

$$n!.$$

In particular, $3! = 3 \cdot 2 \cdot 1 = 6$, as in the preceding example.

Ordered selections

Suppose we select and order k objects from n distinct objects, where $0 \leq k \leq n$.

The successive positions have

$$n, n - 1, \dots, n - k + 1$$

choices.

Hence the number of ordered selections is

$$n(n - 1) \cdots (n - k + 1) = \frac{n!}{(n - k)!}.$$

When $k = n$, this becomes $n!$.

Summary

- ▶ A bijection pairs every element of one set with exactly one element of another set, in both directions.
- ▶ Subsets of an n -element set are in bijection with bit strings of length n .
- ▶ Repeated applications of the product rule count sequences of choices.
- ▶ A permutation is an ordering of distinct objects, and n distinct objects have $n!$ permutations.